

БЧХ-коды и РС-коды

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 $d \leq r+1 = n-k+1$

- 1 —
- 2 код с проверкой на четность
- 3 код Хэмминга
- 4 расширенный код Хэмминга

$$H = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix} \quad (7,4)$$

$$\begin{array}{l} 1 \\ x \\ 1+x \\ x^2 \\ 1+x^2 \\ x^2+x \\ x^2+x+1 \end{array}$$

$GF(2)$

$$1+x+x^3 = P(x)$$

$GF(2^3)$

$$\begin{array}{ll} \alpha^0 & 1 \\ \alpha^1 & x \\ \alpha^2 & x^2 \\ \alpha^3 & 1+x \\ \alpha^4 & x+x^2 \\ \alpha^5 & x^2+x+1 \\ \alpha^6 & 1+x^2 \end{array}$$

$$H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \alpha^3 & \alpha^4 & \alpha^5 & \alpha^6 \\ 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

$$(15, 11) \quad r = n - k = 4$$

$$H = (1 \ \alpha \ \alpha^2 \ \alpha^3 \dots \alpha^{14})$$

$$GF(2) \rightarrow GF(2^4)$$

$$p(x) = 1+x+x^4$$

$$c(x) + e(x) = b(x)$$

$$S = \sum_{i=0}^{n-1} b_i \cdot \alpha^i = \sum_{i=0}^{n-1} (c_i + e_i) \alpha^i =$$

$$= \sum_{i=0}^{n-1} e_i \alpha^i = e(\alpha)$$

В координатах i ошибка $e(x) = x^i$

$$e(x) = x^i + x^j \quad i \neq j$$

$$S = \alpha^i$$

$$S_1 = \alpha^i + \alpha^j$$

$$S_2 = f(\alpha^i) + f(\alpha^j)$$

$$H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \dots & \alpha^{14} \\ \beta_1 & \beta_2 & \dots & \beta_{15} \end{pmatrix} \quad 8 \times 15$$

$\beta_i \in GF(2^m)$

$$\begin{cases} S_1 = \alpha^i + \alpha^j \\ S_2 = f(\alpha^i) + f(\alpha^j) \end{cases}$$

$$H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \alpha^3 & \alpha^4 & \alpha^5 & \alpha^6 & \alpha^7 & \alpha^8 & \alpha^9 & \alpha^{10} \\ 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} & 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} & 1 \\ \alpha^{11} & \alpha^{12} & \alpha^{13} & \alpha^{14} \\ \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} \end{pmatrix}$$

i, j — произвольные сдвиги
 $x = \alpha^i, \quad y = \alpha^j$

$$\begin{cases} x + y = S_1 \\ x^3 + y^3 = S_2 \end{cases} \quad \text{при одной или двух сдвигках } S_1 \neq 0$$

$$\frac{x^3 + y^3}{x + y} = \frac{S_2}{S_1}$$

$$x^3 + y^3 = (x + y)(x^2 + y^2 + xy)$$

$$x^2 + y^2 + xy = \frac{S_2}{S_1}$$

$$(a+b)^2 = a^2 + 2ab + b^2 = a^2 + b^2$$

$$x + y = S_1$$

$$(x + y)^2 = S_1^2$$

$$x^2 + y^2 = S_1^2$$

$$S_1^2 + xy = \frac{S_2}{S_1}$$

$\Rightarrow$

$$xy = \frac{S_2}{S_1} - S_1^2$$

$$\begin{cases} x+y = s_1 \\ xy = \frac{s_2}{s_1} - s_1^2 \end{cases}$$

+ Buemm

$$\begin{cases} z^2 + \sigma_1 z + \sigma_2 = 0 \\ \sigma_1 = s_1 \quad \sigma_2 = \frac{s_2}{s_1} - s_1^2 \end{cases}$$

$$GF(2) \rightarrow GF(2^4) \quad p(x) = 1+x+x^4$$

element d

notation

noe.

	0	0000
1	x	0010
2	x ²	0100
3	x ³	1000
4	1+x	0011
5	x+x ²	0110
6	x ² +x ³	1100
7	1+x+x ³	1011
8	1+x ²	0101
9	x+x ³	1010
10	1+x+x ²	0111
11	x+x ² +x ³	1110
12	1+x+x ² +x ³	1111
13	1+x ² +x ³	1101
14	1+x ³	1001
0	1	0001

$$H = \begin{array}{cccccc} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{array} \rightarrow \begin{array}{cccccc} 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 \end{array}$$

$$\begin{array}{cccccccccccccccc} 1 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \end{array}$$

элементы

$$S_1 = \alpha^2 + \alpha^9 = \alpha^{11}$$

$$S_2 = \alpha^6 + \alpha^{12} = \alpha^4$$

$$\begin{cases} xy = \frac{S_2}{S_1} - S_1^2 \\ x + y = S_1 \end{cases}$$

$$x + y = S_1$$

$$\frac{\alpha^4}{\alpha^{11}} + (\alpha^{11})^2 = \frac{1}{\alpha^7} + \alpha^7 = \alpha^8 + \alpha^7 = \frac{\alpha^{14} + 1}{\alpha^7} =$$

$$= \frac{\alpha^3}{\alpha^7} = \alpha^{-4}$$

$$\sigma_2 = \alpha^{11}, \quad \sigma_1 = \alpha^{11}$$

$$z^2 + \alpha^{11} \cdot z + \alpha^{11} = 0$$

$$z_1 = \alpha^2, \quad z_2 = \alpha^9$$

$$(\alpha^9)^2 + \alpha^{11} \cdot \alpha^9 + \alpha^{11} = \alpha^{18} + \alpha^{20} + \alpha^{11} = \alpha^3 + \alpha^5 + \alpha^{11} = x^3 + x + x^2 + x + x^2 + x^3 = 0$$

$$\begin{aligned} (\alpha^2)^2 + \alpha^{11} \cdot \alpha^2 + \alpha^{11} &= \\ &= \alpha^4 + \alpha^{13} + \alpha^{11} = \\ &= 1 + x + 1 + x^2 + x^3 + x + x^2 + x^3 = \\ &= 0 \end{aligned}$$